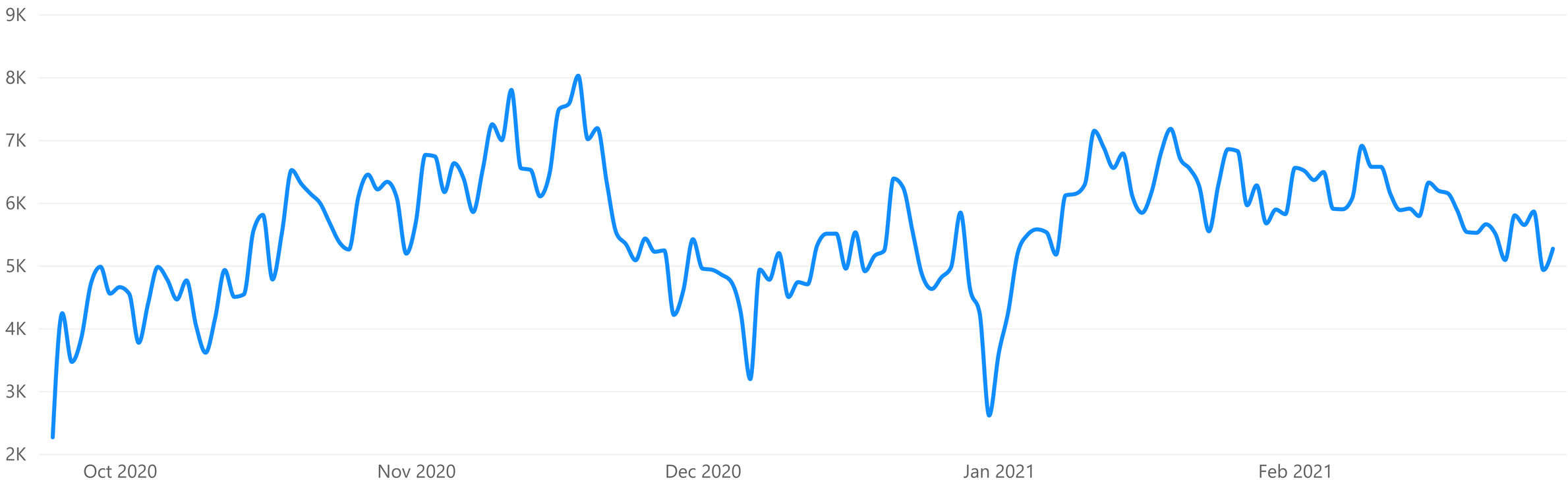
# Customer journeys

REES46 events, September 2020 to February 2021. Recorded purchases are events, not verified orders.

| Retained Events | Observed Users | Observed Products | Observed Sessions |
|---|---|---|---|
| 884K | 407K | 53K | 505K |

## Recorded event volume by date



# Measurement and data quality

Each rate uses its stated population and observation window. Missing brand and category values are retained.

## Rate definitions

| Metric | Definition | Unit | Rate |
|---|---|---|---|
| F01 | Within-session product view-to-cart progression | user-session-product | 7.67% |
| F02 | Within-session ordered product purchase progression | user-session-product | 3.67% |
| F03 | Purchase after cart among view-entered product groups | user-session-product | 47.86% |
| P01 | 7-day observed view-to-cart progression | user-product pair | 7.98% |
| P02 | 7-day ordered view-cart-purchase progression | user-product pair | 3.99% |
| P03 | 7-day observed cart-to-purchase progression | user-product pair | 49.96% |
| S01 | 30-minute session purchase share | reconstructed user-session | 4.86% |
| U01 | Observed-user purchase share | observed user | 5.23% |

## Data quality checks

| Check | Affected rows | Base rows | Share |
|---|---|---|---|
| Exact duplicate extra raw rows | 655 | 885,129 | 0.07% |
| Missing brand in retained events | 212,232 | 884,474 | 24.00% |
| Missing category in retained events | 236,047 | 884,474 | 26.69% |
| Missing supplied session in retained events | 162 | 884,474 | 0.02% |

# Purchase journeys by category

Same user, closed 30-minute session and product. Category differences are observational.

| Groups | Reached cart | Ordered purchases | Purchase rate |
|---|---|---|---|
| 645K | 49K | 24K | 3.67% |

Category

All

## Ordered purchase progression

| Category | Groups | Ordered purchases | Purchase rate |
|---|---|---|---|
| Unknown | 180,978 | 4,778 | 2.64% |
| computers › components › videocards | 72,098 | 4,712 | 6.54% |
| electronics › telephone | 61,107 | 2,652 | 4.34% |
| computers › peripherals › printer | 30,000 | 1,579 | 5.26% |
| stationery › cartrige | 27,113 | 1,679 | 6.19% |
| electronics › audio › acoustic | 20,447 | 486 | 2.38% |
| computers › components › motherboard | 18,540 | 778 | 4.20% |
| computers › notebook | 18,012 | 818 | 4.54% |
| electronics › video › tv | 17,065 | 228 | 1.34% |
| computers › components › cpu | 16,532 | 730 | 4.42% |
| electronics › tablet | 14,192 | 524 | 3.69% |
| auto › accessories › player | 13,092 | 414 | 3.16% |
| computers › components › cooler | 8,111 | 131 | 1.62% |
| appliances › environment › vacuum | 7,015 | 193 | 2.75% |
| construction › tools › drill | 6,717 | 139 | 2.07% |

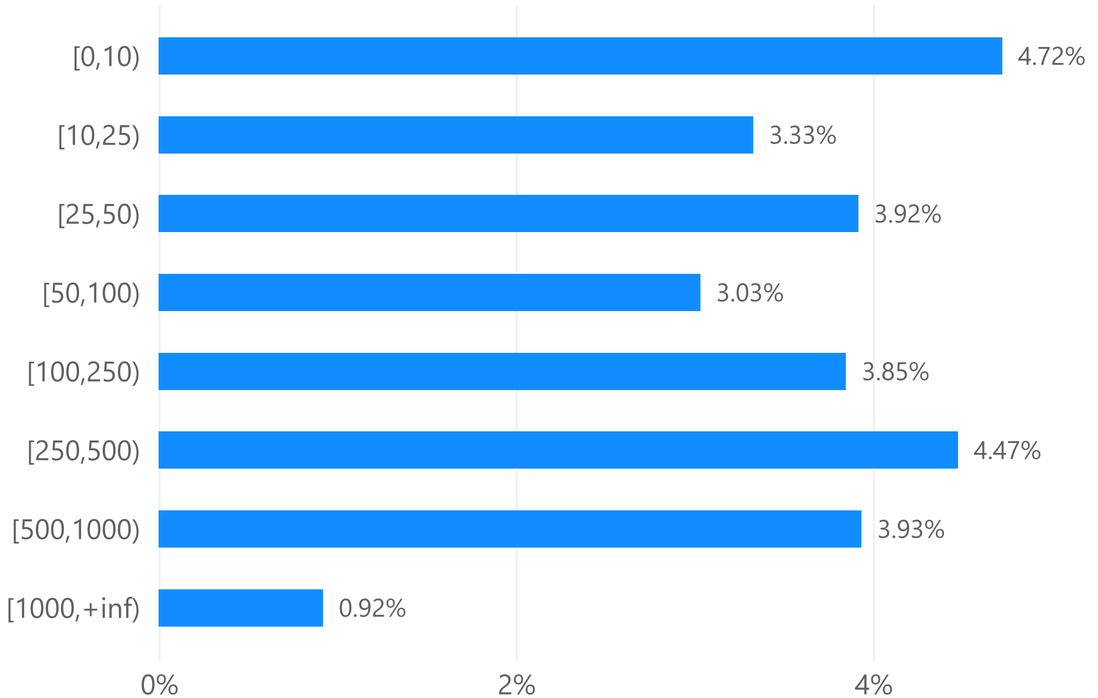
Top 15 of 108 categories by groups. All 108 remain in the full result tables.

# Purchase journeys by price

First-view price in source-reported units; currency is unspecified. Price differences are observational.

| Groups | Reached cart | Ordered purchases | Purchase rate |
|---|---|---|---|
| 645K | 49K | 24K | 3.67% |

## Purchase rate by price band



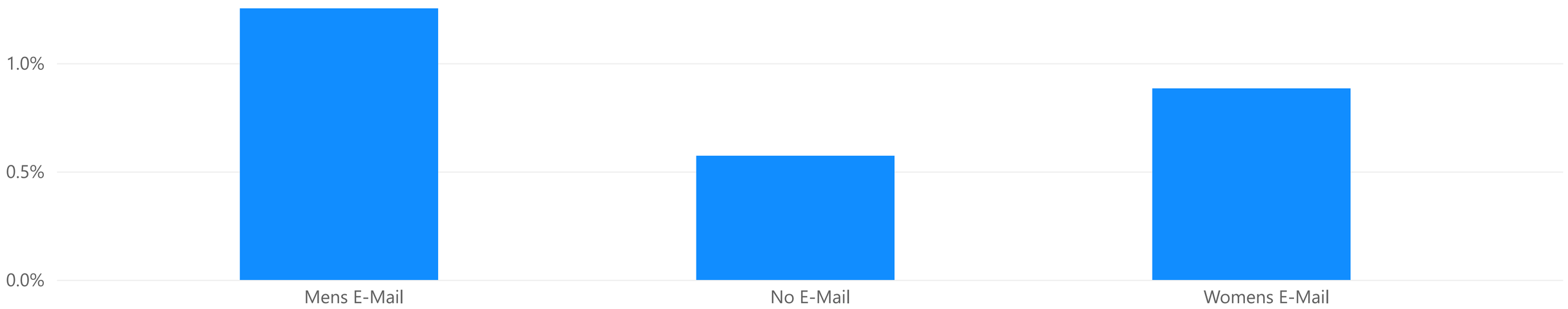

## Groups and ordered purchases

| Price band | Groups | Ordered purchases | Purchase rate |
|---|---|---|---|
| [0,10) | 19,694 | 930 | 4.72% |
| [10,25) | 140,471 | 4,676 | 3.33% |
| [25,50) | 115,609 | 4,528 | 3.92% |
| [50,100) | 127,540 | 3,868 | 3.03% |
| [100,250) | 128,326 | 4,938 | 3.85% |
| [250,500) | 83,230 | 3,724 | 4.47% |
| [500,1000) | 24,347 | 958 | 3.93% |
| [1000,+inf) | 5,443 | 50 | 0.92% |

# The historical email experiment

Hillstrom, 2008. Random assignment to three groups; purchase and visit rates use every assigned customer.

## Purchase rate by assigned email




## All assigned customers

| Arm | Assigned customers | Purchase rate | Visit rate | Mean spend ($) |
|---|---|---|---|---|
| Mens E-Mail | 21,307 | 1.25% | 18.28% | $1.4226 |
| No E-Mail | 21,306 | 0.57% | 10.62% | $0.6528 |
| Womens E-Mail | 21,387 | 0.88% | 15.14% | $1.0772 |

# Email effects against no email

Purchase lift is in percentage points; spend is dollars per assigned customer. Intervals are marginal 95% intervals.

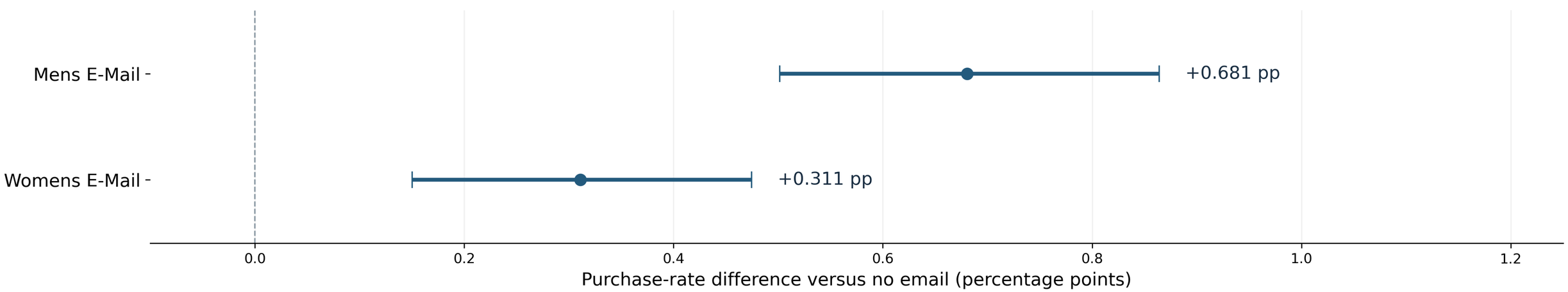
## Purchase effects

| Treatment | Purchase lift (pp) | 95% CI low | 95% CI high | Holm p-value |
|---|---|---|---|---|
| Mens E-Mail | 0.6805 | 0.5014 | 0.8641 | <0.001 |
| Womens E-Mail | 0.3111 | 0.1502 | 0.4745 | <0.001 |

## Exploratory spend effects

| Treatment | Spend lift ($) | 95% CI low ($) | 95% CI high ($) |
|---|---|---|---|
| Mens E-Mail | $0.7698 | $0.4976 | $1.0566 |
| Womens E-Mail | $0.4244 | $0.1703 | $0.6828 |

## Purchase lift and 95% intervals




Marginal Newcombe intervals; the table reports Holm-adjusted p-values. Historical randomized experiment.